

WiDi is not really a new technology though. It was released in 2010 at which time it was limited to a resolution of 720p and was a fairly new software stack which supported only a limited number of devices. There was also a problem with latency being pretty high which seldom caused issues with reception on the receiving device. But now, this is no longer the case and the best part about the newest release is that it has more than 10 times improvement in latency and also supports open standards like Miracast which will enable Miracast receivers to consume output from Intel WiDi chipsets. Apart from this, support has been added for streaming to DLNA receivers. Additionally, wireless support for USB devices has been added so that HID compliant devices can be plugged into the receiver, enabling wireless access for such devices too. Now, it can stream upto Full HD content in 1080p resolution and even 3D games or videos. At the moment, Intel claims that more than 30 million laptops have already been shipped with the WiDi technology.

The latest version of WiDi i.e. 4.0 now adds support for streaming of protected content such as DVDs and Blu-Ray. This is done with the help of HDCP 2.0 output protection technology. It enables availability of even more content across such wireless devices.

INSTALLING WIDI SOFTWARE ON THE DESKTOP

Installing software to run WiDi software and firmware on the supported laptops or ultra-books can be done easily with the help of a one-click installer which makes installation as easy as clicking on the set-up file, accepting the license agreement and you are ready to stream content on WiDi.

Installation (and also uninstallation, for that matter) can also be done in the form of a silent installation. Silent install mode is often used when this installation needs to be automated or to be executed from inside a script. This enables automatic installation/uninstallation of the SDK without manual intervention in the silent install mode. In this mode, we can run the setup.exe file from the command line in a non-verbose mode, with all default options selected and all licenses accepted, and without a prompt for any screens and has all license agreements pre-accepted. This can be done as follows: -

```
Install : setup.exe /s /v/qn
```

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Uninstall : setup.exe /x /s /v/qn
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There is also an option available to enable or disable the Desktop Shortcut. This can be done by providing a 1 or 0 value as follows: -

```
Enable: setup.exe /v"WIDIAPP_DESKTOP_SHORTCUT=1"
```

```
Disable: setup.exe /v"WIDIAPP_DESKTOP_SHORTCUT=0"
```

For further details, help can be accessed from the command line by passing help as a parameter: -

```
Command line: setup.exe /v/help
```

WIDI EXTENSIONS SDK

The WiDi Extensions SDK provides a framework that allows you to develop applications that work and integrate with the Intel Wireless Display technology. It can be used to detect WiDi support, scan for WiDi devices, connect to a device and start the screen display as mirrored or extended version of the current desktop. The SDK comes with a 32-bit COM object and IntelWiDiExtensions.dll bundled with a sample application and its source code.

An important consideration that is to be made, and which has also been specified in the documentation is that while making WiDi apps, the latency of a wireless network needs to be kept in mind. This is because the irregularities in the network should not impact the user experience. Every time a user needs to typically connect to a WiDi display, the device will need to scan for available devices which will then be

made available to start a connection. Now, this is a lengthy and time-consuming process and doing so every time one needs to use WiDi can get frustrating. So, an important workaround is to remember the previously connected device and connect to it either without a scan or to start a scan and connect to it immediately once the device's existence is confirmed.

The prerequisites to building a WiDi application are as follows: -

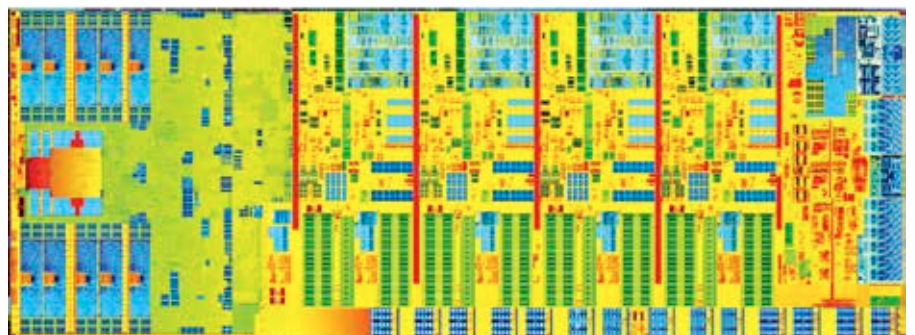
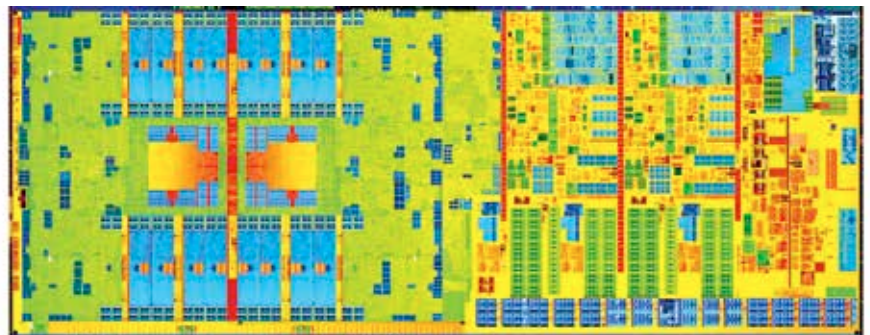
- A WiDi-capable hardware device, since emulation is not possible at the moment.
- Windows 7 Home Premium or Windows 8
- Intel WiDi Software
- Intel WiDi Extension SDK
- Microsoft Visual Studio 2010/2012
- .NET Framework 4.0

The Intel WiDi Extension SDK only supports 32-bit software development in .NET C# and C++ code. Also, it can only be targeted for Windows 8 Desktop apps, since Windows 8 Store apps are not supported. One limitation for developers is that as of now an emulator for WiDi is not available, so applications that need WiDi can be tested only on a WiDi-enabled device.

INSTALLATION OF WIDI EXTENSIONS SDK

Intel WiDi Extensions SDK can be installed on the development system as follows: -

- 1 Intel WiDi Extensions SDK can be downloaded from the following Intel website:



A picture of the die for Intel's dual-core Haswell chips (top) and quad-core offerings (bottom)